ECE-535 FINAL PROJECT REPORT
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1. Introduction

This paper introduces methods to predict the loan defaults better, faster and more accurately by combining advanced preprocessing techniques and using both traditional machine learning and deep learning models. The methods described to predict defaults can be used by financial institutions such as banks, credit unions, digital lenders etc. The model utilizes financial data features i.e. income, experience, house/car ownership etc. to predict the risk flag whether a person will default on the loan or not. The ML models are CPU-friendly and can run on most laptops/personal computers (Intel i3 processors, 4GB RAM, No GPU) while the Deep Learning models are computationally expensive (Intel i5 processor, 8 GB RAM, 4GB NVIDIA GTX 1050 Ti). It is important to note that these are the minimum system requirements, and the performance may vary depending on the hardware used. There might be a few legal restrictions while acquiring datasets from financial institutions which is kept confidential and companies must go through a legal procedure to access the data. However, the authors used publicly available dataset from Kaggle to train and evaluate their models. There are no ethical considerations in the use of these algorithms as there are no personal identifiable features in this dataset. But we need to take care of the fact that the system does fair classification regardless of profession, age, city and state etc.

2. Data Set Collection/Processing:-

The dataset used was originally obtained from Kaggle [1] and it belongs to a Hackathon organized by "Univ.AI". Upon researching the sources, it was found that the dataset was not acquired from an actual financial institution. The dataset originally contains 252,000 samples with 221,004 samples belonging to Class-0 (No Loan Default Risk) and 30,996 belonging to Class-1 (Loan Default Risk). We can calculate the imbalanced ratio using the following equation:

$$Imbalanced\ Ratio = \frac{Total\ Samples\ in\ Majority\ Class}{Total\ Samples\ in\ Minority\ Class} = \frac{221004}{30996} = 7.13$$

So, there is a clear case of Class-0 being dominant in this dataset. Applying machine learning models directly on this dataset would have resulted in flawed model performances regardless of which model is being used. The researchers used SMOTE (Oversampling technique) to create synthetic samples for the minority class (class 1) and Tomek Links (Under sampling technique) to clean noisy or ambiguous data near class boundaries. Using this novel hybrid approach, they modified the original dataset into a more representative and learnable dataset.

The researchers chose preprocessing methods and analyzed feature distributions based on exploratory data analysis (EDA) techniques i.e. heatmaps and distribution charts (Figures 2–5 in the paper). They also examined the correlation between job years and house ownership and class-wise distribution of key features like employment status, loan purpose, etc. The **outliers and noise in dataset** were addressed during data cleaning but the exact method for detecting outliers is not specified. Multiple feature normalization techniques were

discussed but it seems that they used standardization for k-NN and min-max scaling for neural networks. They acknowledged the presence of outliers and noise in the dataset and dealt accordingly. Since the dataset has only 13 features it was easy to decide if there are any irrelevant features.

By identifying the datatypes of feature values i.e. “integer” and “string”, they carefully encoded categorical features to preserve their inherent relationships (especially ordinal variables). The effectiveness of preprocessing was measured by evaluating improvements in model performance metrics such as **accuracy, precision, recall, F1-score, AUC-ROC, and MCC** across both training and testing phases.
To ensure preprocessing did not remove any useful features, the authors avoided unnecessary dimensionality reduction (they only removed ID column from original dataset). However, upon further investigation we found some anomalies in the dataset features and values i.e.

- **90% of customer data (226272) samples belong to unmarried (single) category** and we can think of it being unusual since married people take loans more often, but they consist of only 10% of overall data (25728). The loan default rate for married people is 10% (2636) and unmarried (single) is 12% (28360) samples.
- Another abnormality was the relationship between a person’s age and years of experience. For example, there are some unrealistic examples like a person of age 21 years with 10 years of experience, age 23 with 13 years of experience etc,

Initially, the dataset looked reasonable, but these abnormalities indicate that the dataset doesn’t reflect the values of an actual dataset from a financial institution and seems more like a synthetic dataset. Keeping authenticity aside, we can comment that the applied preprocessing techniques were reasonable and included **outlier removal, feature scaling, categorical encoding, and dimensionality reduction**. The inclusion of **SMOTE-Tomek**, **stratified train-test splits**, and **k-fold cross-validation** aligns with best practices to ensure data quality and model robustness.

One area of improvement could be incorporating **automated outlier detection algorithms** (e.g., Isolation Forest, Local Outlier Factor) [2] for more systematic anomaly handling and **validating preprocessing impacts** using **feature importance metrics** (e.g., Permutations Feature Importance or Random Forest importance scores) [3] before and after transformation to ensure no valuable predictive information was unintentionally lost. Also, the dataset could have been more realistic such as “Deloitte-MachineHack Loan Default Prediction Dataset” [4] or **Lending Club 2007-2020Q3 [5]**

3. Feature Selection

The relevant features for classification were selected through exploratory data analysis (EDA) and domain-driven reasoning. As described in Section III.A of the paper, the authors used visual techniques such as heatmaps (Figure 2) and distribution plots (Figures 3–5) to examine relationships between variables like house ownership, job tenure, and the target

variable (loan risk). Irrelevant features such as the "ID" were manually removed during the dimensionality reduction stage to improve data quality. Categorical variables were encoded using ordinal encoding and one-hot encoding (GetDummies), ensuring that meaningful structure in the data was preserved.
The model performances are related to the selection of features and quality of the dataset. They trained a wide range of ML and Deep Learning models and reported results using metrics such as **accuracy, precision, recall, F1-score, AUC-ROC, and MCC**. The models consistently indicated high performance, especially after applying **SMOTE-Tomek**, showing that the selected features were effective for predicting loan defaults in this dataset.
The data was tabular and therefore, it was not subject to spatial transformations like rotation or scaling (typical in image data).

Yes, there are several additional features that could potentially enhance the predictive performance of the loan defaults i.e. Number of Previous Loans/Defaults, Credit Utilization Rate, Debt-to-Income Ratio (DTI) etc. Some of these features are present in other datasets [4], [5] but not in the dataset used by the authors of this paper. It is understandable that the financial data is kept highly confidential by the institutions and sometimes important financial metrics such as described earlier are not present in most of the publicly available datasets. Moreover, the paper does not employ any feature engineering or integration of data from external sources, which is needed to include complex features like DTI. This shows that they only worked within the bounds of the provided dataset.

4. Machine Learning/Pattern Classification Algorithms:

The authors implemented 7 machine learning algorithms: Gaussian Naive Bayes, Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), AdaBoost, and Gradient Boosting. Additionally, they evaluated 8 deep learning models, including Multi-Layer Perceptron (MLP), Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Transformer, Autoencoder, ResNet, and DenseNet. The problem at hand is a binary classification problem on a structured/tabular financial data. Upon studying the paper thoroughly, we can deduce that the researchers wanted to compare performance of traditional ML and advanced DL models to identify the best suited model with good generalization ability.

The ensemble methods were chosen for their robustness and resistance to overfitting, while the deep learning models were selected for their ability to capture complex, non-linear relationships in structured data. Most of their selections of ML models are justified but we observed that they didn't employ SVM (could provide strong baseline performance for structured data with low dimensionality i.e. 13 features) and Gaussian Mixture Models (could add value in terms of class overlap and interpretability). The Gaussian-NB algorithm assumes feature independence, but features are mostly correlated in real world complex financial datasets. Instead of Ada Boost, a Light GBM model would have provided better boosting performance. The use of algorithms like DenseNet, ResNet is questionable since they are computationally expensive, mainly used for image classification, and the paper

does not explain how tabular features were fed into these models. There was no reasonable justification for the use of autoencoders since they are primary used for anomaly detection and feature reduction. However, they didn't achieve any feature reduction using autoencoders and only removed (ID Column) feature manually.
For parameter selection, they tuned model hyperparameters using techniques such as grid search, random search, and Bayesian optimization. For DL models, they tuned parameters like the number of hidden layers, number of units per layer, activation functions, learning rate, optimizer (e.g., Adam), batch size, and number of training epochs. No specific values were mentioned, but the use of aforementioned strategies seems good enough for hyperparameter tuning. Using the Adam optimizer instead of SGD was a good approach since Adam uses adaptive learning rates, which improves convergence in models like ResNet or LSTM. Even though they used good strategies for tuning, we do have concerns regarding the transparency of parameters used i.e. dropout rates, no. of layers/epochs, batch size, early stopping criteria etc.

From computational point of view, the machine learning models like Naive Bayes and Logistic Regression are lightweight and ready to be deployed on embedded devices. Whereas the deep learning models lie DenseNet and ResNet need significant computational resources, such as GPUs with 8-12GB VRAM which makes them better suited for cluod-based environments. If system is supposed to be deployed in a constrained platform, then simple models like Logistic Regression or Random Forest would make more sense offering reasonable trade-offs between accuracy and efficiency.

5. Performance Evaluation

The paper does a thorough performance evaluation for various Machine Learning and Deep Learning models in a structured manner using six evaluation metrics namely accuracy, precision, recall, F-1 score, AUC-ROC and MCC. These metrics helps to analyze both the general and class wise performance. The paper explores multiple configuration including unbalanced datasets, balanced dataset using SMOTE and SMOTE-Tomek and enhancing models with ensemble methods. Among the ML models Random Forest, Decision Trees and K-Nearest Neighbor performed well with Random Forest achieving nearly perfect score. Among DL models, DenseNet and ResNet performed well, particularly when data balancing techniques were employed along with ensemble methods. Regardless of class imbalances AUC-ROC and MCC provides useful insights into how well the model distinguishes between defaulters and non-defaulters. All these findings were presented visually through charts and tables.

While all the performance metrics discussed were really impressive but there were certain limitations that could impact generalizability. Firstly and most importantly the models were tested on a single dataset that was taken from Kaggle which although had about 252k records but it might not have been able to mirror the variability and noise present in the real-world datasets. Secondly, metrics like AUC-ROC and MCC are useful for imbalanced dataset but the paper does not provide any class-wise breakdown(like confusion matrices

or class-wise precision and recall), which would have been helpful to understand model's performance on class with less amount of data, especially in this case of loan default where false negatives are costly. Also, the evaluation methods does not include any computational considerations like training time, memory usage which are really important for assessing the practicality of deploying models, like the Deep learning model(DenseNet) which require a good amount of computational resources. Another concern is overfitting, because some models have shown 100% accuracy during training which dropped considerably during testing phase which suggest there might be some data leakage or there is no proper regularization. Although the authors did mention importance of interpretability and list SHAP and LIME as tools, they did not present any concrete results or visualizations from the results of these tools. All these points suggest that the evaluation appears to be done thoroughly on the surface but its in-depth and practical relevance could have been improved significantly.

6. Conclusion

This paper introduces an ambitious approach to compare the traditional Machine Learning and modern Deep Learning model for predicting bank loan defaults. It used a publicly available dataset, to which it applied various preprocessing techniques such as SMOTE-Tomek for handling the class imbalance issue. It used a wide range of classification algorithms like Logistic Regression, Random forest etc. and deep learning architecture like MLP, CNN, LSTM etc. The evaluation for the models was carried out by using multiple performance metrics such as accuracy, precision, recall, f1-score, AUC-ROC and MCC.

While the approach covered all the major aspects but there were a few limitations that impacted the validity of the findings. Firstly the dataset is synthetic in nature having anomalies as unrealistic age-exp combination and imbalance in marital status distribution. The preprocessing was done carefully but there was no use of automated outlier detection or deeper validation of feature importance. Secondly the choice of deep learning models like CNN, ResNet and DenseNet was questionable for tabular dataset and that in turn just increased the computational overhead. Thirdly, the absence of configurations of hyperparameter makes it difficult to reproduce the claimed results. Lastly there were a good number of performance metrics, but the authors missed a critical point that is to analyze the trade-offs between them.

In conclusion, the paper makes a valuable contribution by comparing a wide range of models but it lack in terms of dataset quality, justification for model and transparency in evaluation. Further research should take care of dataset quality, appropriate model selection for structured/tabular data and clear reporting of the setup used so that it is easy to reproduce the claimed results.

7. References

[1] https://www.kaggle.com/datasets/subhamjain/loan-prediction-based-on-customer-behavior/data

[2] https://scikit-learn.org/stable/modules/outlier_detection.html

[3] https://machinelearningmastery.com/calculate-feature-importance-with-python/

[4] https://www.kaggle.com/datasets/hemanthsai7/loandefault/code

[5] https://www.kaggle.com/datasets/ethon0426/lending-club-20072020q1/data